Table 4.1 Nutritive value of meat (per 100 g)					
Flesh food	**Energy (Kcal)**	**Protein (g)**	**Fat (g)**	**Calcium (mg)**	**Iron (mg)**
Beef	86	19.4	0.9	3	-
Mutton	194	18.5	13.3	150	2.5

4.1.3 Composition and Nutritive value of meat

Meat has an outstanding nutritive value, contributing substantial amount of high quality proteins and essential minerals and vitamins to the diet. Meat contains 15-20 percent protein of high biological value. The proteins of meat are well utilized by the body, thus ensuring a supply of essential amino acids necessary for growth and maintenance. Meat contains enough iron, phosphorus, zinc and copper to rate as an important source of these minerals.

As far as vitamins are concerned, vitamin A, thiamine and riboflavin are present in liver, kidneys, heart and sweetbreads (the pancreas or the thymus). All lean meats contain thiamine, riboflavin and niacin. Meat is also relatively high in energy value. Meat fats are rich in saturated fatty acids.

4.1.4 Post mortem changes in meat

The changes taking place in meat after slaughter may be grouped under two heads:

1) Onset of rigor mortis
2) Development of tenderness in muscles

Onset of rigor mortis:

After slaughtering, the lean tissues undergo a series of complex physical and chemical changes. As a result, muscles lose their soft pliable nature and become rigid, stiff and inflexible. This is termed as **"rigor mortis".**

Stiff muscle starts to soften and becomes tender when it is held in a cold room temperature between 0°C to 20°C for 1–4 weeks. This is known as **"ripening"** or **"ageing".** During ageing the humidity of the room is to be controlled.

Development of tenderness:

Meat can be made tender by using mechanical methods like pounding, cutting and grinding which break muscle fibre. Addition of salt, vinegar, lime juice and enzymes like papain (raw papaya), bromelin (pineapple) and ficin (figs) also help in tenderizing meat.

4.1.5 Meat Cookery

In the process of cooking, many chemical changes occur affecting the appearance, taste and texture of meat. Cooking meat-

- Develops or improve flavour, colour, aroma.



Fig 4.4: Cooked and raw meat

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